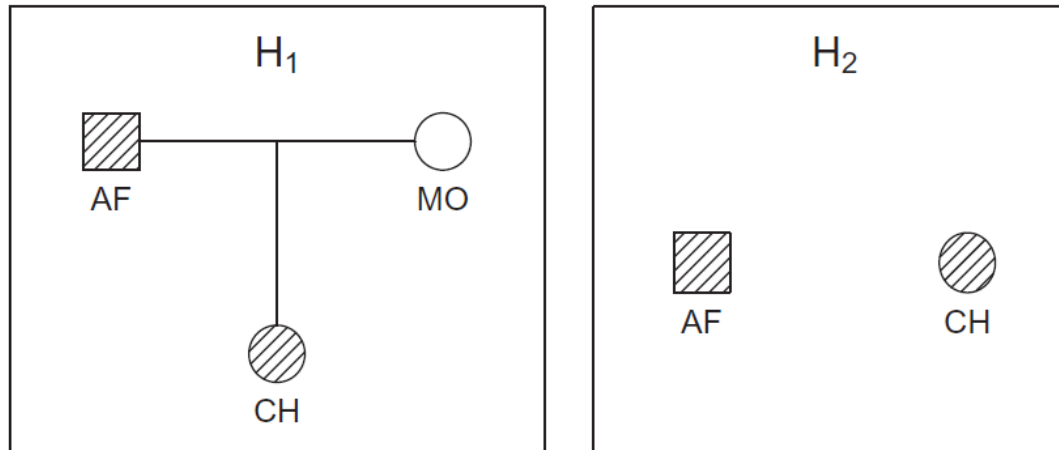




Kinship testing



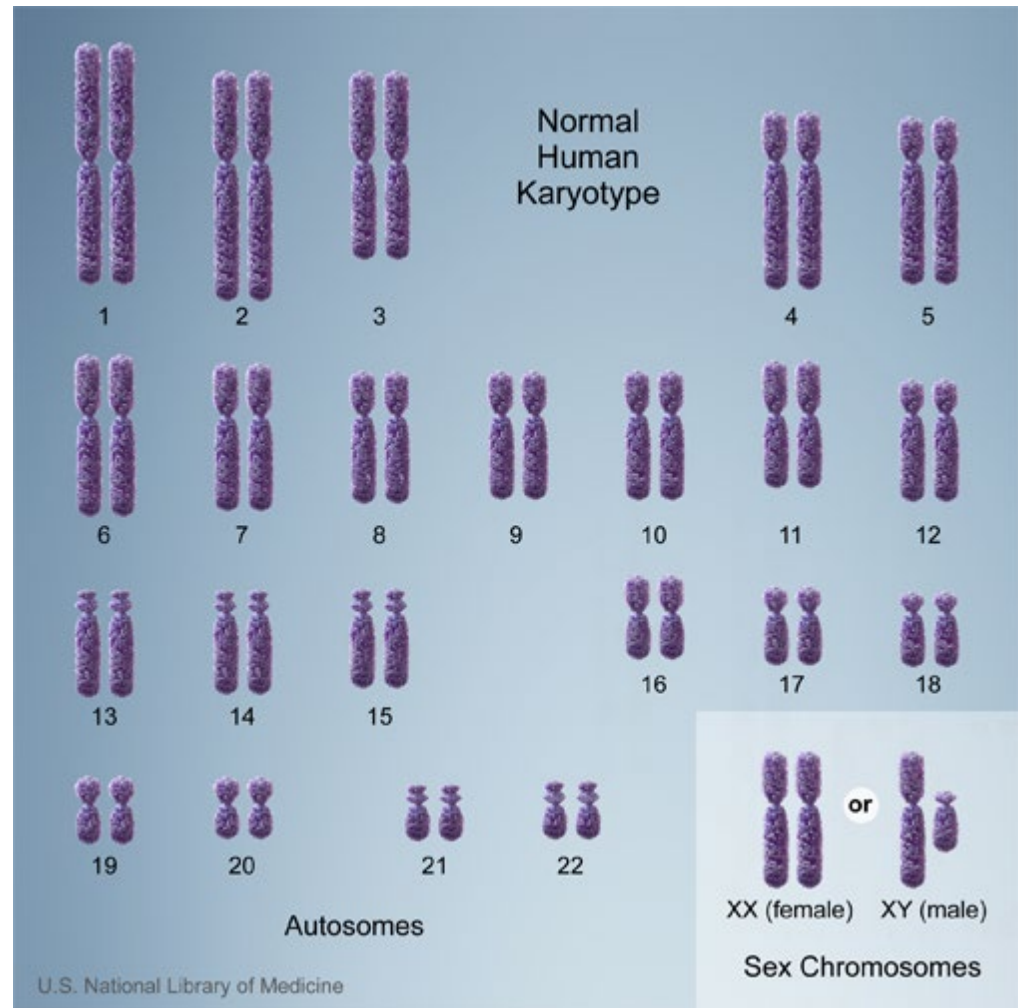
Thore Egeland
Norwegian University of Life Sciences &
Department of Forensic Medicine, Norway

Motivating examples

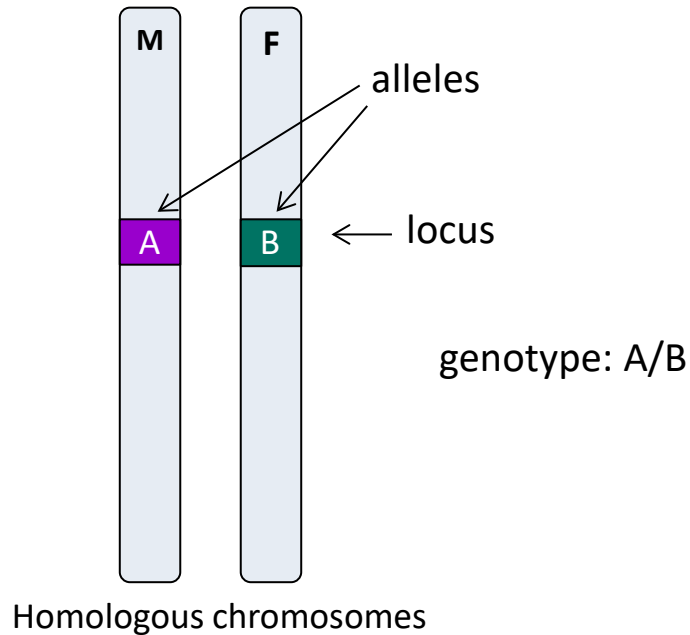
- Kinship testing
 - Close (paternity) or distant (second cousins)
 - Disaster victim identification (DVI)
 - Pedigree reconstruction
 - ...
- We distinguish between
 - *kinship testing*, current topic, where a specific set of alternatives are compared, and
 - *relatedness inference* aiming to find the most probable relationship without restrictions

Genetics terminology

- Locus
- Allele
- Genotype
- Genetic markers
 - SNPs
 - microsatellites



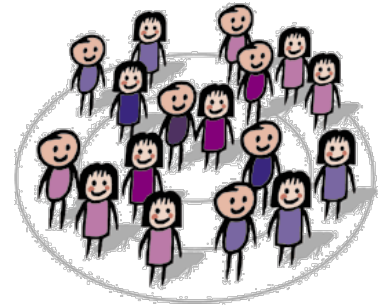
Locus, allele, genotype



- **LOCUS** = a specific place in the genome
- **ALLELE** = any of the alternative forms of a locus
- **GENOTYPE** = the set (usually: pair) of alleles carried at a given locus

Genetic markers

- Small parts of the genome which ...
 - have known position
 - vary in the population
 - are easy to genotype
- SNPs (single nucleotide polymorphisms)
 - two alleles
 - usual requirement: MAF > 1% = minor allele frequency
 - very common in the genome (millions!)
 - used in medical genetics +++
- STRs (short tandem repeats)
 - consecutive repeats of typically 2-5 bases
 - multiallelic: typically 5 - 50 alleles
 - allele names: # repeats
 - used in forensics



...CCGTTA**T**ATGGGC...

...CCGTTA**G**ATGGGC...

...CCGTTA**T**ATGGGC...

...CCGTTA**T**ATGGGC...

...CCGTTA**G**ATGGGC...

...ACG **TTAG** **TTAG** **TTAG** **TTAG** AAC..

...ACG **TTAG** **TTAG** AAC..

...ACG **TTAG** **TTAG** **TTAG** **TTAG** **TTAG** AAC..

Pedigree likelihoods

- Many applications involve probabilities of the following form

A diagram illustrating the components of a probability expression. A light gray rectangular box contains the expression $P(\text{genotypes} \mid \text{pedigree, inheritance model, allele freqs, ...})$. Above the word "data" in the expression is a yellow box containing the word "data". Above the parameter group "inheritance model, allele freqs, ..." is a yellow box containing the symbol Θ . A curly brace is positioned below the parameter group, with a line connecting it to the Θ box.

$$P(\text{genotypes} \mid \text{pedigree, inheritance model, allele freqs, ...})$$

- Often referred to as a *pedigree likelihood*:

A light green rectangular box containing the equation $L(\text{pedigree} \mid \text{data}) = P(\text{data} \mid \text{pedigree}, \Theta)$.

$$L(\text{pedigree} \mid \text{data}) = P(\text{data} \mid \text{pedigree}, \Theta)$$

Three principles of interpretation

1. It is necessary to consider **at least one alternative** proposition
 2. Interpretation is based on questions like:
«What is the probability of the evidence **given** the proposition»
 3. We **condition** on proposition and known circumstances
- These principles lead to the likelihood ratio (LR):

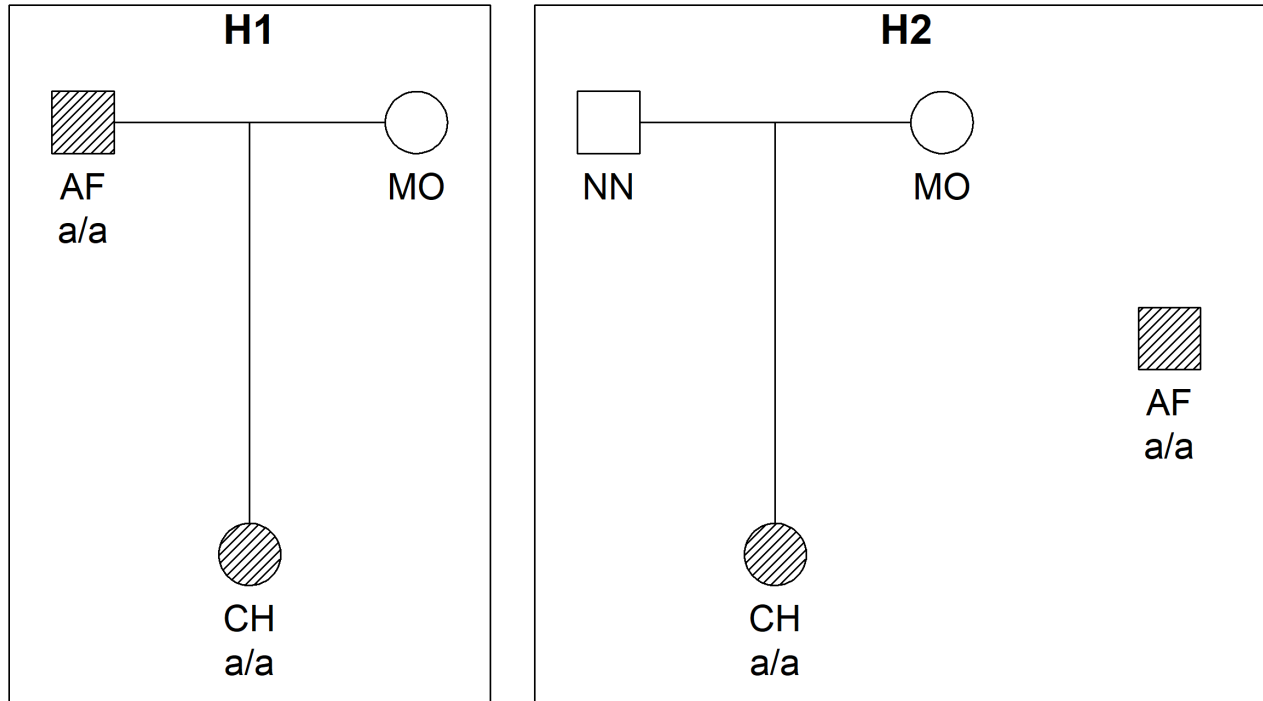
The Likelihood Ratio

- H_1 : The individuals are related according to some pedigree $\mathcal{P}_1$.
- H_2 : The individuals are related according to a different pedigree $\mathcal{P}_2$.

$$\text{LR} = \frac{P(\text{data} \mid H_1, \Theta)}{P(\text{data} \mid H_2, \Theta)}.$$

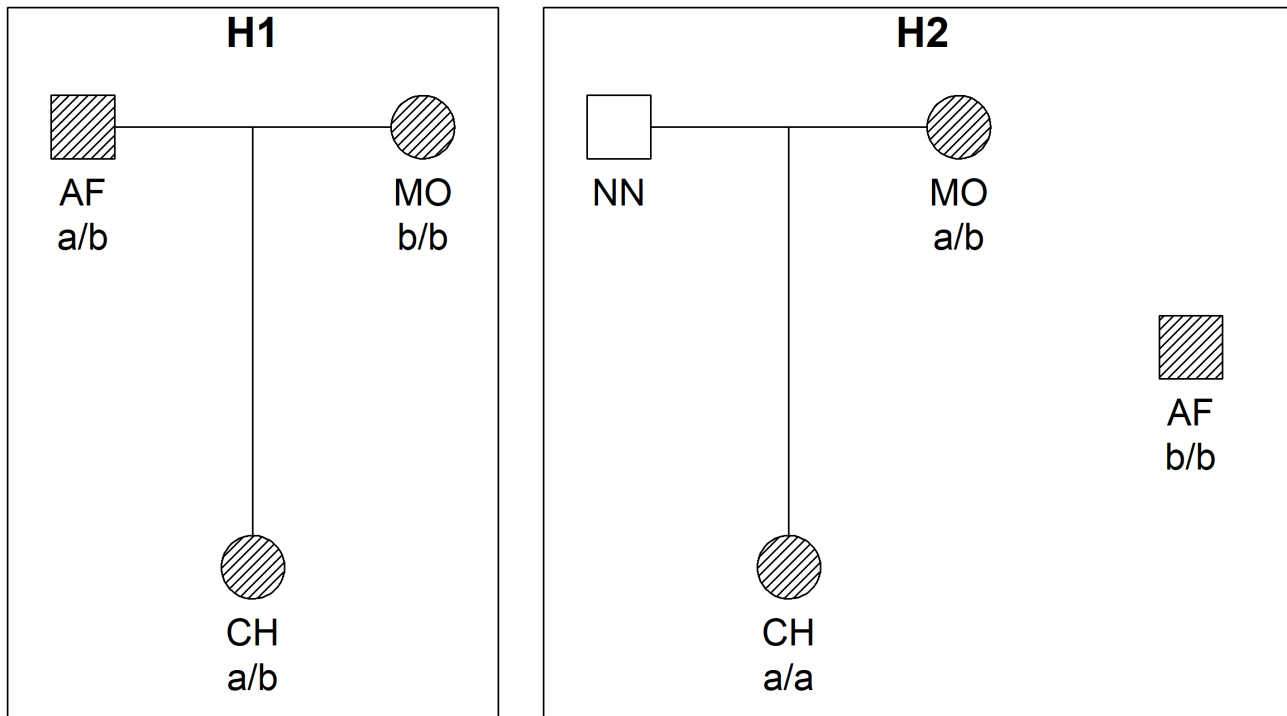
- data: available genotypes
- Θ : fixed model parameters, common to both hypotheses
- **Interpretation:**
 - The LR measures how well H_1 explains the data compared to H_2
- **Default assumptions:**
 - ✓ Hardy Weinberg Equilibrium
 - ✓ **No mutations**
 - ✓ No artefacts (drop out, drop in, genotyping error)
 - ✓ **Independence between markers**

Example 1: Paternity case



$$LR_1 = \frac{P(\text{AF} = a/a, \text{CH} = a/a \mid H_1)}{P(\text{AF} = a/a, \text{CH} = a/a \mid H_2)} = \frac{p_a^2 \cdot p_a}{p_a^2 \cdot p_a^2} = \frac{1}{p_a}.$$

Mother genotyped

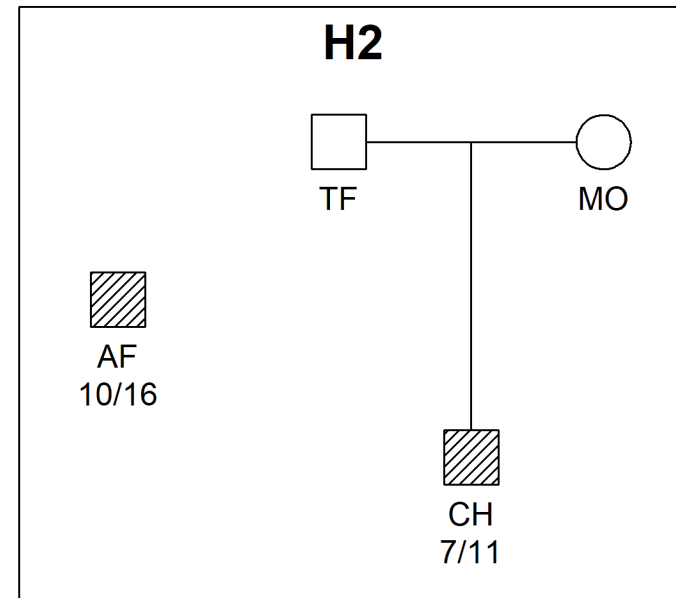
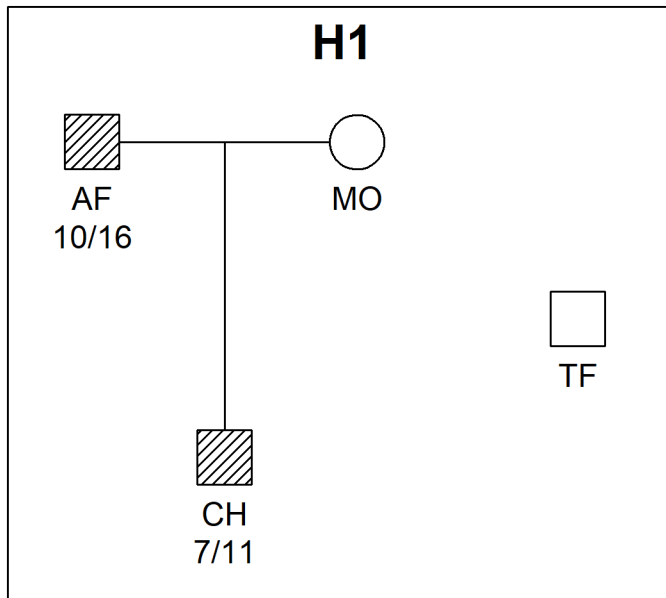


$$LR_2 = \frac{P(\text{AF} = a/b, \text{MO} = b/b, \text{CH} = a/b \mid H_1)}{P(\text{AF} = a/b, \text{MO} = b/b, \text{CH} = a/b \mid H_2)} = \frac{2p_a p_b \cdot p_b^2 \cdot \frac{1}{2}}{2p_a p_b \cdot p_b^2 \cdot p_a} = \frac{1}{2p_a}.$$

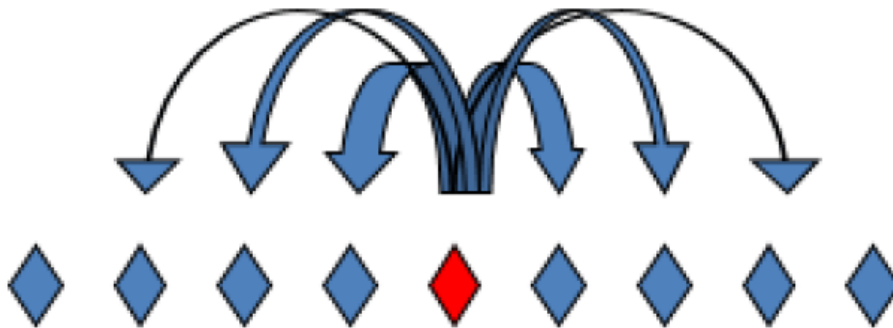
Combined LR

- Assume $p_a = 0.05$ for both markers:
 - $LR_1 = \frac{1}{p_a} = 20$
 - $LR_2 = \frac{1}{2p_a} = 10$
- Assuming independence:
 - $LR = LR_1 \cdot LR_2 = 20 \cdot 10 = 200$
- **Interpretation:**
The data is 200 times more likely if we assume H_1 to be true rather than H_2

Mutation?

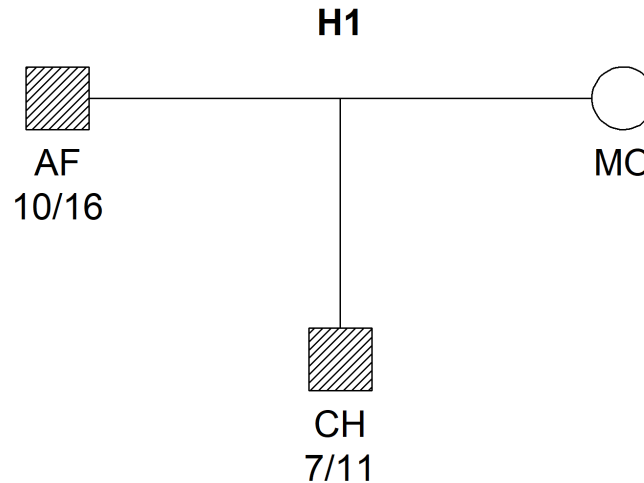


Mutation models



- ▶ Mutation rates higher in males.
- ▶ Short mutations more likely: One step mutation more likely than two steps and so on.
- ▶ Mutation rates:
<http://www.cstl.nist.gov/strbase/mutation.htm>

Dealing with mutations



Strategies for handling mutations

- Exclude inconsistent markers from the analysis. **Not recommended**
- Apply mutation modelling only to inconsistent markers
- Apply mutation modelling to *all* markers. **Recommended**

Kinship testing with linked markers

Forensic Science International: Genetics 86 (2027) 103578



Contents lists available at [ScienceDirect](#)

Forensic Science International: Genetics

journal homepage: www.elsevier.com/locate/fsigen

Research Paper

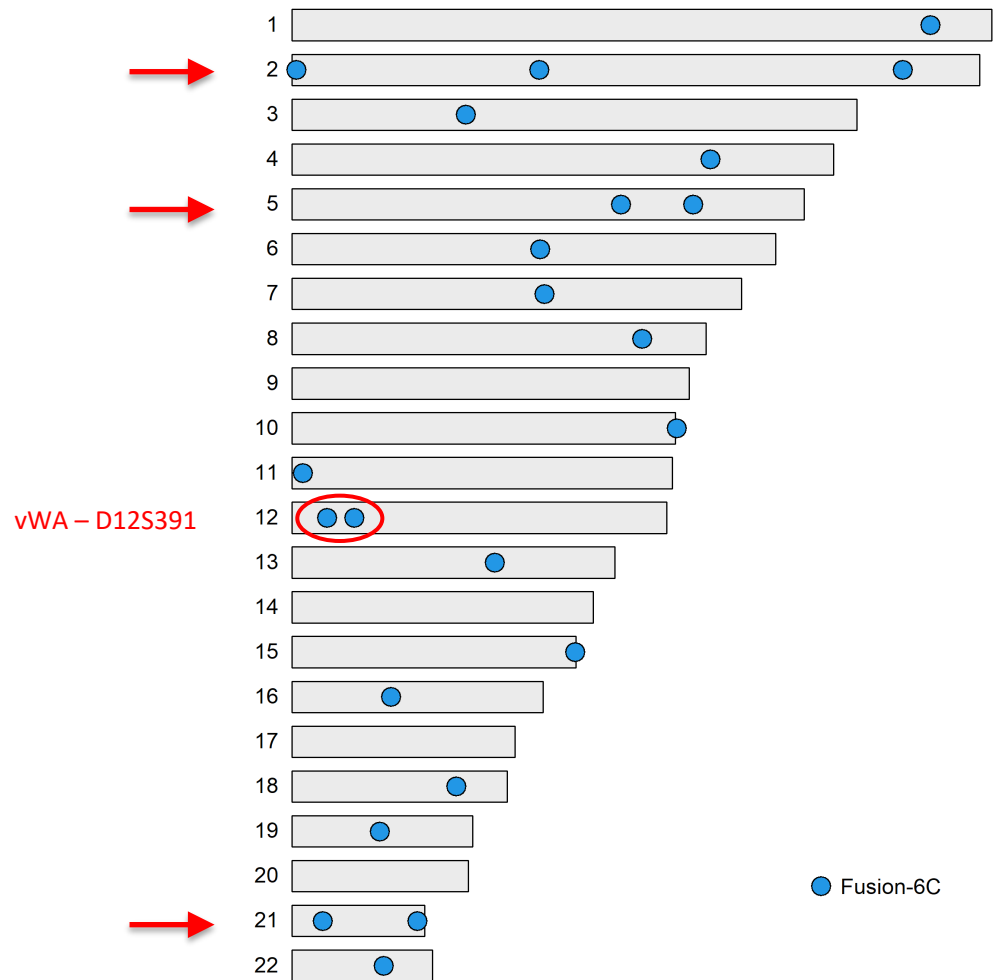
KLINK: A program for kinship testing with pairwise linked STR markers

Magnus D. Vigeland *, Siv Gilfillan

Department of Forensic Sciences, Oslo University Hospital, Oslo, 0424, Norway

PowerPlex® Fusion 6C

- 27 autosomal STR markers
- 1 close pair (< 25 cM)
- 4 linkage groups

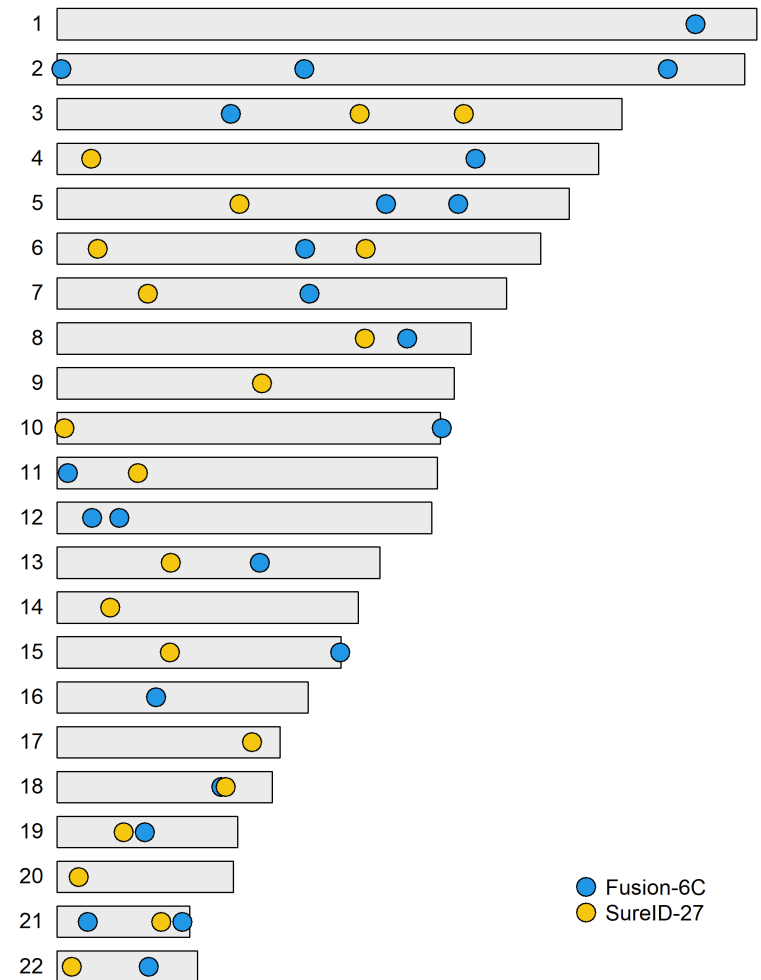


PowerPlex® Fusion 6C

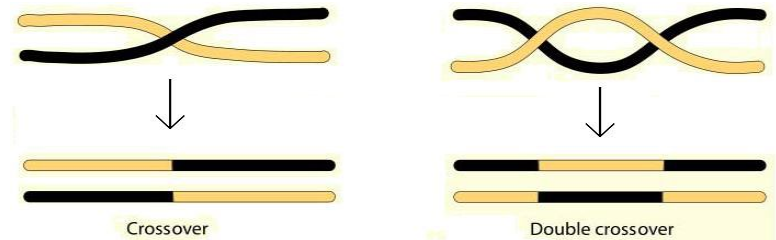
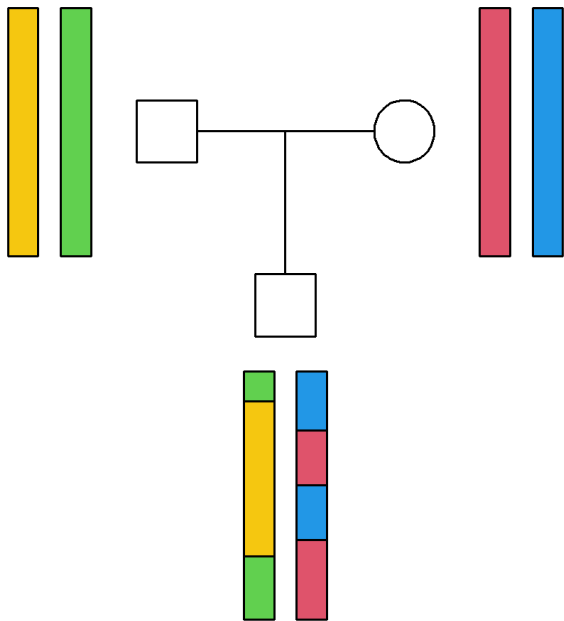
- 27 autosomal STR markers
- 1 close pair (< 25 cM)
- 4 linkage groups

Fusion 6C + SureID® 27

- 49 markers
- 6 close pairs (< 25 cM)
- 16 linkage groups

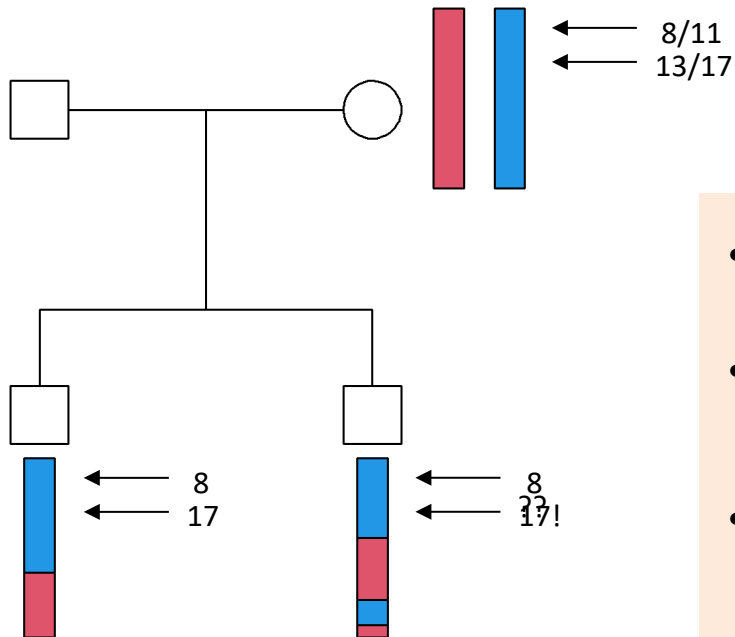


Meiotic recombination



- **Genetic distance** between two loci:
= average crossovers per meiosis
- Units:
 - 1 Morgan (M) = 1 crossover per meiosis
 - 1 centiMorgan (cM) = 0.01 M
- The human genome: Ca 30 Morgan

Why do we care about linkage?



- Linkage → dependent inheritance
- Irrelevant in paternity cases (duo/trio)
(unless *phased* data)
- Relevant if ≥ 2 informative,
non-independent meioses

Software handling linked markers



MERLIN

Pros

- **Gold standard** for linkage
- Arbitrary linkage

Cons

- Command-line based
- Max 30 alleles
- No mutation modelling

FamLink2

Pros

- Designed for forensics
- Good with SNPs
- GUI + many features
- *Quick analysis* mode

Cons

- No mut. models in QA

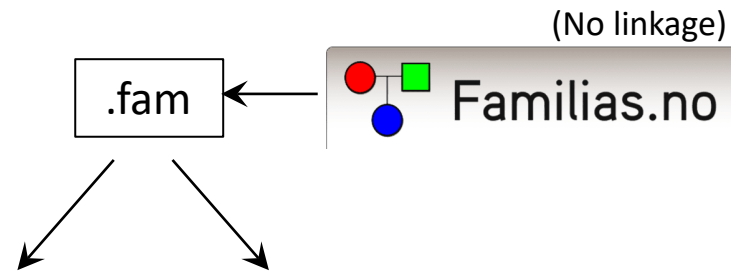
KLINK

Pros

- Designed for kinship STRs
- GUI with visualisations
- Any mutation models
- Tailor-made Excel reports

Cons

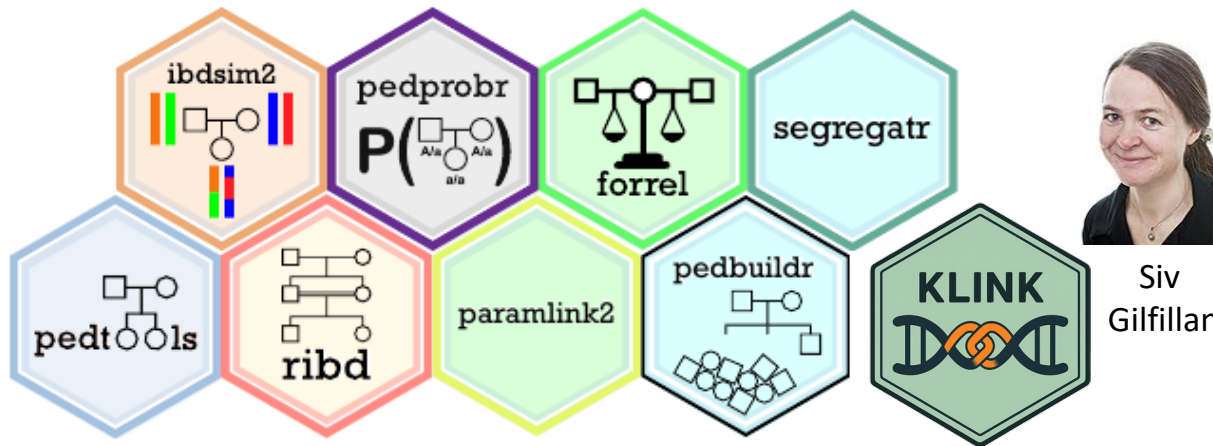
- Only pairwise linkage



Home page: <https://magnusdv.github.io/pedsuite>
Source code: <https://github.com/magnusdv>

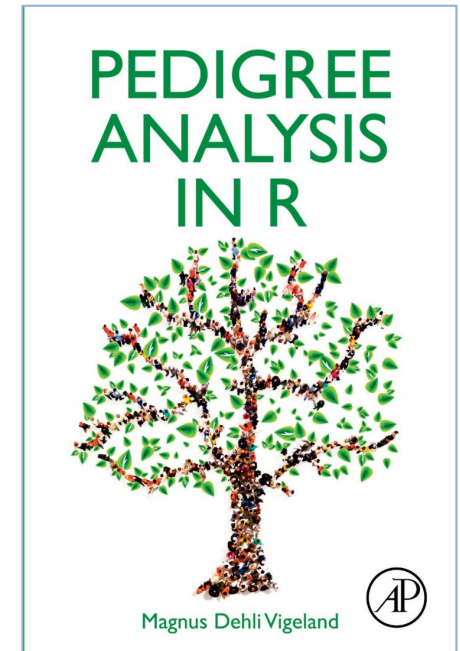
The **pedsuite**:

A collection of packages for pedigree analysis in R



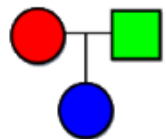
KLINK is available in two ways:

- From R: `KLINK::launchApp()`
- Online: <https://magnusdv.shinyapps.io/klink/>
- Input. Familias file:



Familias

- New homepage merging familias.name into <https://familias.no/>
- New maintainer, developer: Franco Marsico



Familias - Tutorial

By: Daniel Kling, Lourdes Prieto,
Franco Marsico and Thore Egeland



Petter Mostad

KLINK: Kinship with pairwise linked markers



INPUT

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Load .fam file

Upload complete

Marker map

☒ Built-in ☐ Custom

(Optional) .xml

Example 1

Example 2

SETTINGS

Map function

☒ Kosambi ☐ Haldane

Empty markers

☒ Hide ☐ Show

Likelihoods

☒ Hide ☐ Show ☐ Loglik

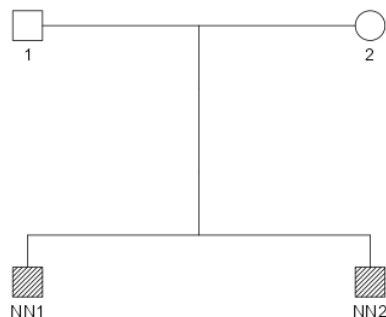
Decimals

Ignore linkage above (cM)

Calculate LR

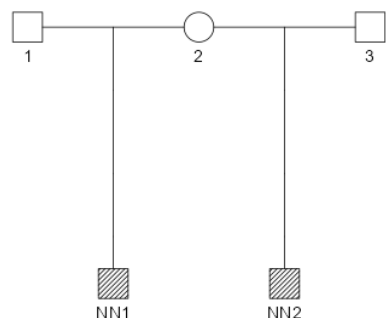
Ped 1

☒ Hide untyped components



Ped 2

Marker



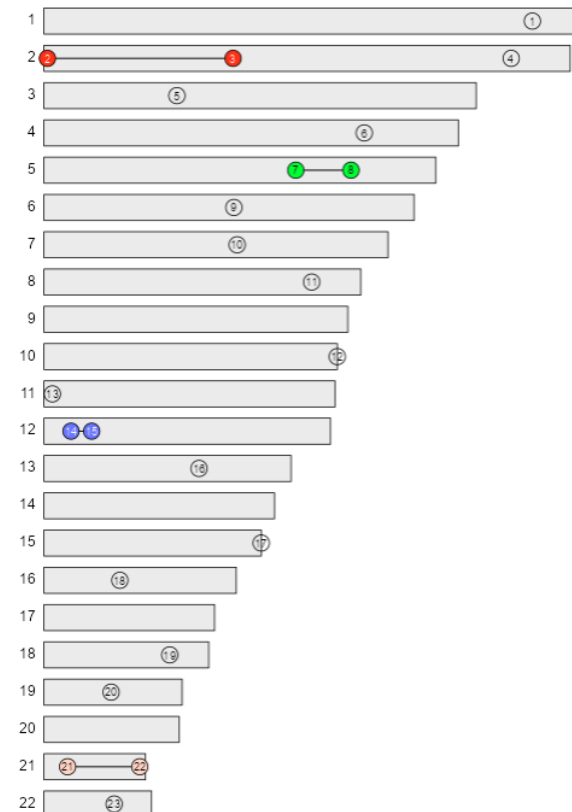
Linkage map

Marker data

LR table

Download

Marker	Chr	cM	Kit
1 D1S1656	1	244.235	Fusion6C/SureID
2 TPOX	2	1.666	Fusion6C
3 D2S441	2	90.479	Fusion6C/SureID
4 D2S1338	2	223.483	Fusion6C
5 D3S1358	3	67.179	Fusion6C
6 FGA	4	156.813	Fusion6C
7 D5S818	5	126.673	Fusion6C
8 CSF1PO	5	154.434	Fusion6C
9 SE33	6	95.449	Fusion6C
10 D7S820	7	100.201	Fusion6C
11 D8S1179	8	136.443	Fusion6C
12 D10S1248	10	169.899	Fusion6C/SureID
13 TH01	11	4.489	Fusion6C
14 vWA	12	15.630	Fusion6C
15 D12S391	12	27.571	Fusion6C/SureID
16 D13S317	13	79.831	Fusion6C
17 Penta E	15	124.051	Fusion6C
18 D16S539	16	50.000	Fusion6C/SureID
19 D18S51	18	88.921	Fusion6C
20 D19S433	19	51.726	Fusion6C
21 D21S11	21	14.646	Fusion6C
22 Penta D	21	59.376	Fusion6C
23 D22S1045	22	46.214	Fusion6C/SureID



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KLINK: Kinship with pairwise linked markers



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Example 2

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☒ Hide ☐ Show ☐ Loglik

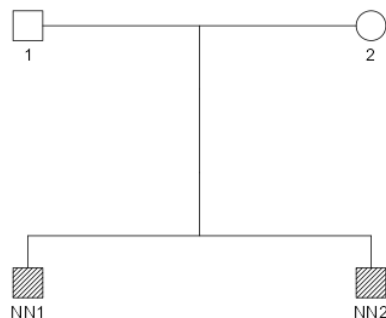
Decimals

Ignore linkage above (cM)

Calculate LR

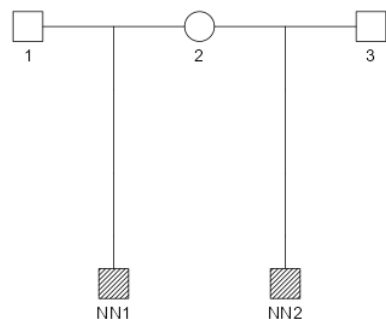
Ped 1

☒ Hide untyped components



Ped 2

Marker



Linkage map

Marker data

LR table

Download

Marker	Person1	Person2	Alleles	PIC	MinFreq	Model	Rate	Rate2	Range	Stat	Lump
D1S1656	15.3/16	15/15.3	17	0.891	0.001	Stepwise	0.001/0.002	1e-06	0.1	No	No
TPOX	8/9	8/8	9	0.569	0.001	Stepwise	0.001/0.002	1e-06	0.1	No	No
D2S441	10/14	10/14	13	0.696	0.001	Stepwise	0.001/0.002	1e-06	0.1	No	No
D2S1338	17/20	24/24	13	0.863	0.001	Stepwise	0.001/0.002	1e-06	0.1	No	No
D3S1358	14/18	14/18	12	0.766	0.001	Stepwise	0.001/0.002	1e-06	0.1	No	No
FGA	20/21	20/21	25	0.855	0.001	Stepwise	0.001/0.002	1e-06	0.1	No	No
D5S818	9/12	9/12	9	0.645	0.001	Stepwise	0.001/0.002	1e-06	0.1	No	No
CSF1PO	13/13	13/13	11	0.688	0.001	Stepwise	0.001/0.002	1e-06	0.1	No	No
SE33	25.2/28.2	20/21	55	0.947	0.001	Stepwise	0.001/0.002	1e-06	0.1	No	No
D7S820	8/12	8/12	19	0.794	0.001	Stepwise	0.001/0.002	1e-06	0.1	No	No
D8S1179	14/14	14/14	12	0.779	0.001	Stepwise	0.001/0.002	1e-06	0.1	No	No
D10S1248	13/14	13/16	9	0.718	0.001	Stepwise	0.001/0.002	1e-06	0.1	No	No
TH01	6/9.3	6/9.3	10	0.734	0.001	Stepwise	0.001/0.002	1e-06	0.1	No	No
vWA	14/20	14/20	12	0.775	0.001	Stepwise	0.001/0.002	1e-06	0.1	No	No
D12S391	18/19.3	19.3/21	23	0.884	0.001	Stepwise	0.001/0.002	1e-06	0.1	No	No
D13S317	11/12	12/12	9	0.779	0.001	Stepwise	0.001/0.002	1e-06	0.1	No	No
Penta E	7/11	11/16	21	0.890	0.001	Stepwise	0.001/0.002	1e-06	0.1	No	No
D16S539	11/11	11/11	9	0.737	0.001	Stepwise	0.001/0.002	1e-06	0.1	No	No
D18S51	12/15	12/15	23	0.866	0.001	Stepwise	0.001/0.002	1e-06	0.1	No	No
D19S433	12/13	12/13	17	0.739	0.001	Stepwise	0.001/0.002	1e-06	0.1	No	No
D21S11	27/28	27/31	26	0.822	0.001	Stepwise	0.001/0.002	1e-06	0.1	No	No
Penta D	10/12	10/12	24	0.807	0.001	Stepwise	0.001/0.002	1e-06	0.1	No	No
D22S1045	11/15	15/16	9	0.674	0.0012	Stepwise	0.001/0.002	1e-06	0.1	No	No

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KLINK: Kinship with pairwise linked markers



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Example 1

Example 2

SETTINGS

Map function

☒ Kosambi ☐ Haldane

Empty markers

☐ Hide ☒ Show

Likelihoods

☐ Hide ☒ Show ☐ Loglik

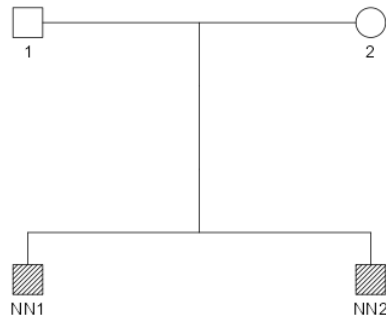
Decimals

Ignore linkage above (cM)

Calculate LR

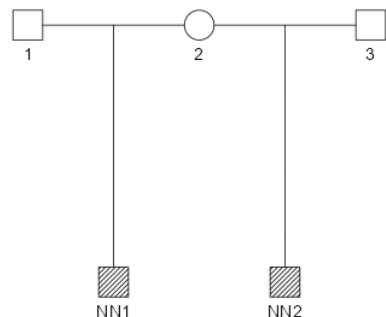
Ped 1

☒ Hide untyped components



Ped 2

Marker



Linkage map

Marker data

LR table

Marker	Person1	Person2	LR	Linked	No link	No mut
D1S1656	15.3/16	15/15.3	0.882	0.882	0.882	0.874
TPOX	8/9	8/8	0.742	1.720	1.722	1.713
D2S441	10/14	10/14	2.322			
D2S1338	17/20	24/24	0.500	0.500	0.500	0.500
D3S1358	14/18	14/18	3.787	3.787	3.787	3.799
FGA	20/21	20/21	3.192	3.192	3.192	3.202
D5S818	9/12	9/12	3.205	23.903	21.215	24.158
CSF1PO	13/13	13/13	6.618			
SE33	25.2/28.2	20/21	0.500	0.500	0.500	0.500
D7S820	8/12	8/12	3.448	3.448	3.448	3.457
D8S1179	14/14	14/14	2.813	2.813	2.813	2.822
D10S1248	13/14	13/16	0.721	0.721	0.721	0.721
TH01	6/9.3	6/9.3	2.011	2.011	2.011	2.012
vWA	14/20	14/20	9.499	4.531	9.353	4.532
D12S391	18/19.3	19.3/21	0.985			
D13S317	11/12	12/12	0.819	0.819	0.819	0.816
Penta E	7/11	11/16	0.857	0.857	0.857	0.858
D16S539	11/11	11/11	2.129	2.129	2.129	2.133
D18S51	12/15	12/15	3.724	3.724	3.724	3.735
D19S433	12/13	12/13	3.705	3.705	3.705	3.722
D21S11	27/28	27/31	0.932	2.674	2.780	2.680
Penta D	10/12	10/12	2.984			
D22S1045	11/15	15/16	0.714	0.714	0.714	0.714
Total LR				2.75×10^5	5.25×10^5	2.80×10^5

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KLINK: Kinship with pairwise linked markers



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(Optional) .xml

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☐ Kosambi ☒ Haldane

Empty markers

☐ Hide ☒ Show

Likelihoods

☐ Hide ☒ Show ☐ Loglik

Decimals

Ignore linkage above (cM)

Calculate LR

Ped 1



Ped 2



Hide untyped

KLINK-sibship.xlsx									
File Home Insert Page Layout Formulas Data Review View Automate Help Acrobat									
A1 : X Y fx Resultatrapport: sibship									
A	B	C	D	E	F	G	H	I	
1	Resultatrapport: sibship								
2	Markør	Person1	Person2	LR				Samples	
3	1	D1S1656	15.3-16	15-15.3	0.882			Person1	NN1
4	2	D2S1338	17-20	24-24	0.500			Person2	NN2
5	3	D3S1358	14-18	14-18	3.787				
6	4	FGA	20-21	20-21	3.192				
7	5	SE33	25.2-28.2	20-21	0.500			Relationship	
8	6	D7S820	8-12	8-12	3.448			Ped 1	Full siblings
9	7	D8S1179	14-14	14-14	2.813			Ped 2	Half siblings
10	8	D10S1248	13-14	13-16	0.721				
11	9	TH01	6-9.3	6-9.3	2.011				
12	10	D13S317	11-12	12-12	0.819			Notifications	
13	11	Penta E	7-11	11-16	0.857			No notifications recorded	
14	12	D16S539	11-11	11-11	2.129				
15	13	D18S51	12-15	12-15	3.724			Settings	
16	14	D19S433	12-13	12-13	3.705			1	KLINK version: 1.1.0
17	15	D22S1045	11-15	15-16	0.714			2	Familias file: sibship.fam
18	16	TPOX	8-9	8-8	1.720	0.742		3	Genetic map: BUILT-IN
19	17	D2S441	10-14	10-14		2.322		4	Database: NorskDB_2023
20	18	D5S818	9-12	9-12	23.903	3.205		5	Map function: Kosambi
21	19	CSF1PO	13-13	13-13		6.618		6	Max distance: 1000 cM
22	20	vWA	14-20	14-20	4.531	9.499			
23	21	D12S391	18-19.3	19.3-21		0.985			
24	22	D21S11	27-28	27-31	2.674	0.932			
25	23	Penta D	10-12	10-12		2.984			
26	Total LR				2.752e+05				
27									
28									
29									

<

>

Linkage map

Marker data

LR table

Notifications

Full report

Plots

Unlinked report

Download

No mut

0.874

1.713

0.500

3.799

3.202

24.158

0.500

3.457

2.822

0.721

2.012

4.532

0.816

0.858

2.133

3.735

3.722

2.680

0.714

2.80×10^5

This is KLINK version 1.1.0 ([changelog](#) | [official releases](#)). If you encounter problems, please file a [bug report](#). See also the [KLINK homepage](#) for more information.

KLINK: Kinship with pairwise linked markers

INPUT RESET ?

Familias .fam file

No file selected

Optional .xml file

No file selected

Example 1 Example 2

SETTINGS ?

Marker map

☒ Built-in ☐ Custom

Map function

☒ Kosambi ☐ Haldane

Mutation model

☒ Original ☐ Simple ☐ Off

Empty markers

☒ Hide ☐ Show

Likelihoods

☒ Hide ☐ Show ☐ Loglik

Decimals **Unlinked > cM**

3

Most options can be left untouched

Click on question marks to see explanations

Mutation modelling:

- "Original" = use models in the .fam file (if any)
- "Simple" = Equal model with rate 0.001 (all markers)